

MLOps – Scenario-Based Case Study Assignment

Course: Machine Learning Operations (MLOps)

Assignment Type: Case Study / Scenario-Based

Submission Deadline: 10 September 2026, 5:00 PM

Total Marks: 07

Mode: Individual

1. Assignment Objective

The objective of this assignment is to understand how Machine Learning models are developed, deployed, monitored, and maintained in real-world production environments.

Students are required to analyze a given business scenario and design an appropriate **MLOps workflow/pipeline** for solving the problem.

The assignment is inspired by real-world MLOps use cases such as fraud detection, demand forecasting, predictive maintenance, recommendation systems, customer-support automation, anomaly detection, churn prediction, and computer-vision quality inspection.

2. Case Study Scenarios

Choose **ONE** of the following scenarios.

Scenario 1: Online Payment Fraud Detection

A digital payment company processes thousands of transactions every minute. The company wants to develop an ML model that identifies potentially fraudulent transactions in real time.

The model should classify a transaction as:

- Genuine
- Suspicious
- Fraudulent

However, fraud patterns change frequently, and a model that performs well today may become less accurate over time.

Your Task:

Design an MLOps solution that explains:

1. How transaction data will be collected.
 2. How data will be preprocessed.
 3. How the fraud detection model will be trained.
 4. How the model will be deployed for real-time prediction.
 5. How model performance will be monitored.
 6. How the model will be retrained when fraud patterns change.
 7. How model versions will be managed.
 8. How a failed model deployment can be rolled back.
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Scenario 2: Retail Demand Forecasting

An e-commerce company wants to predict the demand for different products for the next 7 days.

The company has historical information about:

- Product sales
- Price
- Discounts
- Festivals
- Seasons
- Location
- Customer demand

The company wants to reduce both **stock-outs** and **excess inventory**.

Your Task:

Design an MLOps pipeline for the demand forecasting system.

Your solution should explain:

1. Data collection and storage.
 2. Data preprocessing and feature engineering.
 3. Model training.
 4. Model evaluation.
 5. Batch prediction.
 6. Model deployment.
 7. Automated retraining.
 8. Monitoring of prediction accuracy.
 9. Handling of data drift.
 10. Important business KPIs.
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Scenario 3: Predictive Maintenance in Manufacturing

A manufacturing company has machines equipped with sensors that continuously record:

- Temperature
- Vibration
- Pressure
- Operating hours
- Machine speed

The company wants to predict whether a machine is likely to fail within the next 24 hours.

Unexpected machine failures result in significant production losses.

Your Task:

Design an end-to-end MLOps solution.

Explain:

1. How sensor data will be collected.
2. How time-series data will be processed.
3. What features can be extracted.
4. Which ML model you would consider.
5. How the model will be trained and evaluated.
6. Whether you would use edge or cloud deployment and why.
7. How model performance will be monitored.
8. How new failure data will be used for retraining.
9. How model updates will be deployed safely.
10. What KPIs should be monitored.

Scenario 4: Customer Churn Prediction

A telecom company wants to identify customers who are likely to discontinue their subscription.

Available data includes:

- Customer age
- Subscription plan
- Monthly bill
- Number of complaints
- Call/data usage
- Payment history
- Customer service interactions

The company wants to identify high-risk customers and offer them suitable retention schemes.

Your Task:

Design an MLOps pipeline for customer churn prediction.

Your solution should cover:

1. Data collection.
2. Data preprocessing.
3. Feature engineering.
4. Model training.
5. Model evaluation.
6. Model deployment.
7. Batch prediction.
8. Monitoring.
9. Retraining strategy.
10. A/B testing of retention campaigns.

Scenario 5: Product Recommendation System

An online shopping platform wants to recommend relevant products to customers based on their browsing and purchasing behavior.

The system should consider:

- Previous purchases
- Product views
- Search history
- Ratings
- Customer preferences
- Similar products

The recommendation system needs to continuously learn from new customer interactions.

Your Task:

Design an MLOps architecture for this recommendation system.

Discuss:

1. Data collection.
2. Feature engineering.
3. Model training.
4. Model evaluation.
5. Recommendation serving.
6. Real-time vs batch processing.
7. Model deployment.
8. A/B testing.
9. Monitoring.
10. Retraining and model versioning.

Scenario 6: Computer Vision Quality Inspection

A manufacturing company wants to automatically detect defective products using cameras installed on its production line.

The ML model should classify products as:

- Good
- Defective

However, lighting conditions, camera positions, and product designs may change over time.

Your Task:

Design an MLOps pipeline for the computer vision system.

Explain:

1. Image data collection.
2. Image preprocessing.
3. Dataset labeling.
4. Model training.
5. Model evaluation.
6. Model deployment.
7. Monitoring.
8. Handling data/model drift.
9. Retraining.
10. Safe deployment of updated models.

3. Common Requirements for All Scenarios

Your case study must include the following components:

A. Problem Definition

Clearly explain:

- What is the business problem?
- Why is ML required?
- Who will use the prediction?
- What is the expected outcome?

B. Data Pipeline

Explain:

Data Source → Data Collection → Data Storage → Data Preprocessing → Feature Engineering

C. ML Pipeline

Explain:

Training Data → Model Training → Model Evaluation → Model Selection → Model Registry

D. Deployment

Explain how the trained model will be deployed.

You may consider:

- REST API
- Batch deployment
- Real-time deployment
- Edge deployment
- Cloud deployment
- Docker/containerization

E. Monitoring

Identify what should be monitored after deployment.

For example:

- Model accuracy
- Precision
- Recall
- Latency
- Data drift
- Prediction drift
- System performance
- Business KPI

F. Retraining

Explain:

- When should the model be retrained?
- What new data will be used?
- How will the new model be evaluated?
- Who/what will approve the new model?

G. Versioning and Rollback

Explain how you would maintain:

- Dataset versions
- Code versions
- Model versions
- Pipeline versions

Also explain what happens if the newly deployed model performs poorly.

4. Mandatory Architecture Diagram

Each student/group must create an **end-to-end MLOps architecture diagram**.

The diagram should contain, wherever applicable:

```
Data Sources
  ↓
Data Ingestion
  ↓
Data Storage
  ↓
Data Preprocessing
  ↓
Feature Engineering
  ↓
Model Training
  ↓
Model Evaluation
  ↓
Model Registry
  ↓
CI/CD Pipeline
  ↓
Model Deployment
  ↓
Prediction
  ↓
Monitoring
  ↓
Feedback / New Data
  ↓
Model Retraining
```

Students should modify the architecture according to their selected scenario.

5. Tools and Technologies

Students may propose suitable MLOps tools such as:

- Python
- Scikit-learn
- TensorFlow / PyTorch
- Git / GitHub
- Docker
- MLflow
- FastAPI / Flask
- Apache Airflow
- Kubernetes
- AWS / Microsoft Azure / Google Cloud
- Prometheus / Grafana
- DVC

Note: Students do not need to use every tool. They should justify why a particular tool is appropriate for their selected scenario.

6. Expected Deliverables

Each student/group must submit:

1. Case Study Report

Recommended length: **5–8 pages**

The report should contain:

1. Title
2. Problem Statement
3. Business Scenario
4. Data Description
5. Proposed ML Approach
6. MLOps Pipeline
7. Deployment Strategy
8. Monitoring Strategy
9. Retraining Strategy
10. Versioning and Rollback
11. Architecture Diagram
12. Expected Challenges
13. Conclusion
14. References

2. Architecture Diagram

A clear end-to-end MLOps workflow diagram must be included.

3. Optional Implementation

Students may implement a small prototype using Python, MLflow, Docker, FastAPI, GitHub, or other suitable tools.

Implementation is encouraged but the primary focus of this assignment is the MLOps design and case-study analysis.

7. Important Instructions

1. Select **only ONE scenario**.
 2. The solution should focus on **MLOps**, not only on ML model development.
 3. Clearly explain why your proposed deployment and MLOps strategy is suitable.
 4. Include an **end-to-end architecture diagram**.
 5. Avoid copying the reference article directly. Your solution must be **your own analysis and design**.
 6. Clearly mention all external sources used.
 7. AI tools may be used for learning and brainstorming, but students must understand and explain their submitted work.
 8. Ensure that the report is properly formatted and submitted before the deadline.
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Submission Deadline

10 September 2026, 5:00 PM

Late submissions may be subject to the course/instructor's late-submission policy.

Reference

Students may refer to the following article for understanding real-world MLOps use cases:

“MLOps Use Cases: 12 Practical Examples Teams Run in Production” – DEV Community

[Reference Article – MLOps Use Cases](#)

The reference discusses how production ML systems require more than model training, including versioning, CI/CD, deployment, monitoring, feedback loops, governance, and retraining.